

scGPT Human Validation — Reviewer 1 (Blinded)

Purpose

Independently determine how each citing paper uses scGPT. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Calibration: complete CAL-01 to CAL-03 independently, then discuss those three cases and freeze the codebook.

Formal validation: complete every HV row in the displayed order. Do not open the other reviewer document or the consensus document until both formal reviews are locked.

Reconciliation: after both reviewers lock their answers, the coordinator copies both records into the consensus document. Discuss only disagreements. Use `REFERENCE_UNRESOLVED` when agreement is not possible.

Evidence boundary: use only the linked approved article versions. Search all occurrences of “scGPT” and inspect the surrounding methods, results, figures, tables, and supplements that are linked.

Allowed values

Citation validity: Yes / No / Unclear

Method executed: Yes / No / Unclear

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located

Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.

Review table

Article and approved evidence links	Reviewer 1 independent record
CAL-01 CALIBRATION scMGCL: accurate and efficient integration representation of single-cell multi-omics data 2025 Bioinformatics DOI: 10.1093/bioinformatics/btaf392 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: we applied scGPT to generate embeddings for the PBMC Multiome dataset and concatenated them with the original features as input to scMGCL for multi-omics integration Notes:
CAL-02 CALIBRATION Genome language modeling (GLM): a beginner's cheat sheet 2025 Biology Methods and Protocols DOI: 10.1093/biomethods/bpaf022 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: ...are being pre-trained on genomic sequences and biomedical literature whereas, scGPT [69] models single cell RNA expression profiles. Notes:

CAL-03 CALIBRATION Building a learnable universal coordinate system for single-cell atlas with a joint-VAE model 2024 Communications Biology DOI: 10.1038/s42003-024-06564-0 Article page Full text 1 Full text 2	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Furthermore, as an interpretable model, UniCoord is complemented with the fashionable large-scale pretrained models^{42,43,44,45}. The results produced by these large models can be interpreted by UniCoord, and UniCoord can generate pseudo cells with high confidence to fit the huge demand of data for large model training. Notes:
HV-042 FORMAL VALIDATION Foundation model reveals the shared organization of transcription and topologically associating domains 2026 Cell Systems DOI: 10.1016/j.cels.2026.101675 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Derived from scGPT (single-cell Generative Pre-trained Transformer), a foundation model trained on 33 million cells, CTS captures whether genes occupy similar roles within cellular programs. Notes:
HV-045 FORMAL VALIDATION Opportunities for artificial intelligence and synthetic biology in designing living drug delivery systems 2026 Advanced Drug Delivery Reviews DOI: 10.1016/j.addr.2026.115883 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: For the first time, researchers are leveraging large language models to form “foundation models” that integrate sequence information at the DNA, RNA, and/or protein levels to predict biochemical, cellular, and/or molecular function [57], [58], [59], [60], [61], [62]. Notes:

HV-027 FORMAL VALIDATION Bioinformatics Copilot 2.0 for Transcriptomic Data Analysis 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.08.15.607673 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Additionally, foundational models such as scFoundation7 and scGPT8, pretrained on extensive single-cell transcriptomic datasets, are being used to facilitate cell type annotation, perturbation prediction, and multi-omic integration. Notes:
HV-044 FORMAL VALIDATION CellAgent: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.05.13.593861 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:To evaluate the performance of CellAgent in batch correction for single-cell transcriptomics data, we compared CellAgent with scVI [7], Scanorama [32], Harmony [33], CellPLM [34], scGPT [10], and Combat [35] on five datasets [36]. Notes:
HV-013 FORMAL VALIDATION Integrating In silico perturbation with multilayer omics to decode regulatory networks in cancer immunity: a new frontier in precision oncology 2026 Frontiers in Immunology DOI: 10.3389/fimmu.2026.1857447 Article page Full text 1	Review status:LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:While early deep learning frameworks provided a starting point, current trends involve models like scGPT and GearNet, which utilize pre-trained cell embeddings to learn the language of gene expression. Notes:

HV-049 FORMAL VALIDATION Routine FFPE sections support clinically compatible single-nucleus transcriptomics across six human cancer types 2026 DOI: 10.64898/2026.07.23.740343 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Extend/Apply to biological data Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:To enable prospective sample-by-sample analysis, data were processed using OneMap, that leverages scGPT23 foundation-model embeddings to classify cellular populations from individual samples. Notes:
HV-008 FORMAL VALIDATION Towards Cross-Sample Alignment for Multi-Modal Representation Learning in Spatial Transcriptomics 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.03.02.709002 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:To address some of these limitations, recent transcriptomics foundation models have been developed to generate richer non-linear embeddings that capture subtle transcriptomics patterns Cui et al. (2024a;b); Hao et al. (2024). Notes:
HV-016 FORMAL VALIDATION Investigation of cell development and tissue structure network based on natural Language processing of scRNA-seq data 2025 Journal of Translational Medicine DOI: 10.1186/s12967-025-06263-2 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:While more advanced contextual embedding methods based on Transformer architectures, such as scBERT [31], Geneformer [32], and scGPT [33], have emerged in recent years, they typically require higher computational resources and more complex model structures. Notes:

HV-035 FORMAL VALIDATION Revealing the Best Strategies for Rare Cell Type Detection in Multi-Sample Single-Cell Datasets 2025 Genes DOI: 10.3390/genes17010031 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We first use the pretrained scGPT model to transform each cell's gene expression profile into a latent embedding, and then apply Isolation Forest to these embeddings to compute anomaly (rare-cell) scores. Notes:
HV-050 FORMAL VALIDATION Context-dependent utility and robustness of pretrained single-cell foundation model representations across analytical tasks 2026 DOI: 10.64898/2026.06.18.733285 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency: Sufficient Supporting quote + locator:scGPT ranked within the top five in five of the six evaluated metrics and achieved the highest score in kBET, demonstrating broad competitiveness across integration criteria. Notes:
HV-046 FORMAL VALIDATION AUPRC: a metric for evaluating the performance of in-silico perturbation methods in identifying differentially expressed genes 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf426 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Most recently, works that incorporate transformer based large language models like scGPT [22] leverage generative pre-training to learn biological embeddings for predicting genetic perturbation responses, annotating cell types, and integrating multi-omic datasets. Notes:

HV-001 FORMAL VALIDATION Generative genomics accurately predicts future experimental results 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.09.08.674753 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We compared to embeddings from scGPT19, a widely used, pretrained model of scRNA-seq, for a reference point. Notes:
HV-003 FORMAL VALIDATION sciLaMA: A Single-Cell Representation Learning Framework to Leverage Prior Knowledge from Large Language Models 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.01.28.635153 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:To evaluate sciLaMA's performance and assess the impact of incorporating prior knowledge encoded as gene embeddings on cell-level tasks (Section 4), we benchmarked sciLaMA against the state-of-the-art (SOTA) model scVI (Lopez et al., 2018) and foundation models such as scGPT, CellPLM, and GenePT (Chen & Zou, 2024; Cui et al., 2024; Wen et al., 2023). Notes:
HV-031 FORMAL VALIDATION C5aR1 inhibition reprograms tumor associated macrophages and reverses PARP inhibitor resistance in breast cancer 2024 Nature Communications DOI: 10.1038/s41467-024-48637-y Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:For further clarification of macrophage phenotypes, a reverse annotation was done by scGPT with MDST datasets as a training model and a large human myeloid dataset as a test data45 (See Methods). Notes:

HV-043 FORMAL VALIDATION DECIPHER for learning disentangled cellular embeddings in large-scale heterogeneous spatial omics data 2025 Nature Communications DOI: 10.1038/s41467-025-63140-8 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Embedding, or the procedure of representing original high-dimensional spatial data as low-dimensional vectors, has been widely adopted to enrich biologically relevant signals while simultaneously removing technical artifacts^{6,7}. Notes:
HV-007 FORMAL VALIDATION Benchmarking gene expression reconstruction from single-cell latent representations 2026 DOI: 10.64898/2026.06.15.731445 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:In foundation-model reconstruction, we evaluate embeddings learned from four foundation models (SE embedding from STATE1, scConcept, scGPT, SCimilarity)^{17,24–26} paired with three decoder architectures (multilayer perceptron (MLP), k-nearest neighbors (KNN), and Transformer). Notes:
HV-011 FORMAL VALIDATION NanoCellAnnotator: Formalizing Expert Cell Type Annotation with Large Language Models 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.21.728965 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Furthermore, many LLM-based annotation pipelines depend on large cloud-hosted models, limiting reproducibility and practical deployment in clinical or privacy-sensitive settings [4] [5]. Notes:

HV-002 FORMAL VALIDATION PertDiffBench: Benchmarking Diffusion Models for Single-Cell Perturbation Response Prediction 2026 DOI: 10.64898/2026.06.13.732013 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We compare performance across feature dimensions and evaluate how different single-cell data representation methods work with the DDPM backbone, including STATE, scFoundation, scGPT, Tahoe-x1, Geneformer, Scimilarity, and scVI. Notes:
HV-021 FORMAL VALIDATION scaLR: a low-resource deep neural network-based platform for single cell analysis and biomarker discovery 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf243 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Recently, deep neural network (DNN) based models have been introduced to enhance the accuracy and scalability of cell type annotation [5–7]. Notes:
HV-020 FORMAL VALIDATION Challenges in AI-driven Biomedical Multimodal Data Fusion and Analysis 2025 Genomics Proteomics & Bioinformatics DOI: 10.1093/gpbjnl/qzaf011 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:scGPT [14] focuses on processing these high-dimensional and sparse RNA data to capture complex relationships between cells, thereby supporting transfer learning across datasets and even across species. Notes:

HV-009 FORMAL VALIDATION iPro-CSAF: identification of promoters based on convolutional spiking neural networks and spiking attention mechanism 2025 PeerJ Computer Science DOI: 10.7717/peerj-cs.2761 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:In bioinformatics fields researchers have developed many machine learning methods and especially deep learning methods to help solving problems in many fields including gene expression (Al taweraqi & King, 2022), protein structures (AlQuraishi, 2021; Tubiana, Schneidman-Duhovny & Wolfson, 2022) genomic (Dalla-Torre et al., 2024) and single-cell biology (Cui et al., 2024), etc. Notes:
HV-006 FORMAL VALIDATION Artificial Intelligence agents for biological research: a survey 2026 Briefings in Bioinformatics DOI: 10.1093/bib/bbag075 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:The survey clarifies how biological AI agents differ from foundation models (e.g. scGPT and scFoundation) by enabling multi-step planning, contextual reasoning, workflow refinement, and tool-driven analysis. Notes:

HV-025 FORMAL VALIDATION Hybrid Attention-Enhanced Regularized Convolutional Autoencoder for Robust EEG Feature Extraction in Seizure Prediction 2025 IEEE Access DOI: 10.1109/access.2025.3608302 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Moreover, feature quality was evaluated based on biological conservation and technical batch correction, consistent with several clustering performance metrics adopted in a previous study [45]. Notes:
HV-037 FORMAL VALIDATION Tahoe-x1: Scaling Perturbation-Trained Single-Cell Foundation Models to 3 Billion Parameters 2025 DOI: 10.1101/2025.10.23.683759 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We follow a similar expression binning strategy as was proposed in scGPT [2]. Notes:
HV-028 FORMAL VALIDATION Single-cell sequencing in precision management of chronic myeloid leukemia 2026 Journal of Translational Genetics and Genomics DOI: 10.20517/jtgg.2026.42 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Foundation models such as scGPT further illustrate how large-scale pretrained representations may support cell-type annotation, perturbation modeling, and cross-dataset transfer learning, although their reliability in rare hematologic malignant states still requires benchmarking and interpretability assessment[65]. Notes:

HV-012 FORMAL VALIDATION MOSAIC: Intra-tumoral heterogeneity characterization through large-scale spatial and cell-resolved multi-omics profiling 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.05.15.654189 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Last but not least, another encouraging approach towards multimodal AI is the development of foundational models for biology, for which MOSAIC data offers an unprecedented training opportunity. Indeed, recent advances have demonstrated the potential of foundational models for histology images (Filiot et al. 2024), single-cell transcriptomics (Cui et al. 2024), but also multimodal data including spatial transcriptomics (Lin et al. 2024). Notes:
HV-017 FORMAL VALIDATION CellMemory: hierarchical interpretation of out-of-distribution cells using bottlenecked transformer 2025 Genome biology DOI: 10.1186/s13059-025-03638-y Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Compared to advanced methods such as scGPT and scPoli, the cell embeddings generated by CellMemory exhibited clear consistency with high-resolution cell states (Fig. 2f). Notes:

HV-040 FORMAL VALIDATION Clifti-GPT: Privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data 2025 DOI: 10.21203/rs.3.rs-7917089/v1 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Leveraging recent advances in foundation models (FMs) for scRNA-seq data, which have demonstrated strong generalization capabilities across diverse datasets, we developed Clifti-GPT, built upon scGPT FM, as a privacy-preserving federated workflow that enables multiple institutions to collaboratively adapt a pretrained FM without sharing raw data. Notes:
HV-022 FORMAL VALIDATION PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.26.707870 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Apply to biological data/Extend Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:To illustrate the router's utility in a biologically meaningful application, we used PantheonOS to perform in-silico gene perturbation analysis on the human fetal heart single-cell data using scGPT43, which was automatically selected by the router for this perturbation task. Notes:

HV-005 FORMAL VALIDATION A multimodal cell-free RNA language model for liquid biopsy applications 2025 Nature Machine Intelligence DOI: 10.1038/s42256-025-01148-x Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Single-cell foundation models have also made prominent strides in transcriptomics. Models such as scBERT19, Exceiver20, GeneCompass21, scFoundation22, Geneformer23, scGPT24 and UCE25 have demonstrated efficacy in tasks such as cell-type classification and gene expression imputation. Although these models have shown promise, their full potential in genomics is still being realized26,27,28. Notes:
HV-023 FORMAL VALIDATION Genetic Convergence Analysis of CRISPR Perturbations Deciphers Gene Functional Similarity 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.11.13.688060 Article page Full text 1 Full text 2	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Generative models have been developed to predict the outcome of perturbation experiments that have not been performed [52, 53, 54]. [45] envision these pseudo-data as one step toward producing a perturbation cell and tissue atlas. Notes:

HV-026 FORMAL VALIDATION Cancer-associated fibroblasts enhance colorectal cancer lymphatic metastasis via CLEC11A/LGR5-mediated WNT pathway activation 2025 Journal of Clinical Investigation DOI: 10.1172/jci194243 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:Inspired by the capabilities of foundation models in single-cell representation, particularly their exceptional performance in cross-task transfer and few-shot learning, here we applied a pretrained single-cell model, scGPT (35), to generate cell embeddings. These LLM-derived embeddings serve as reliable feature inputs for constructing machine-learning classifiers, particularly for low-confidence cells. Notes:
HV-015 FORMAL VALIDATION scGPT-spatial: Continual Pretraining of Single-Cell Foundation Model for Spatial Transcriptomics 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.02.05.636714 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Extend Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We introduce scGPT-spatial, a specialized foundation model for spatial transcriptomics continually pretrained on our previously published scGPT scRNA-seq foundation model. Notes:

HV-024 FORMAL VALIDATION Scouter predicts transcriptional responses to genetic perturbations with large language model embeddings 2025 Nature Computational Science DOI: 10.1038/s43588-025-00912-8 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:However, given the growing interest in adapting gene expression foundation models for this task, we also conducted additional experiments to benchmark Scouter against several such models—scGPT9, scELMo17 and scFoundation10—which were pretrained by their authors on large collections of scRNA-seq datasets and fine-tuned by us for perturbation prediction. Notes:
HV-038 FORMAL VALIDATION cspray: Distributed Single Cell Transcriptome Analysis 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.06.704110 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:On the first point, one may consider foundation models for single cells (see e.g. [9–12]) to be a perfect fit for cell type labeling across many samples. Notes:

HV-041 FORMAL VALIDATION Nicheformer: a foundation model for single-cell and spatial omics 2025 Nature Methods DOI: 10.1038/s41592-025-02814-z Article page Full text 1 Full text 2	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We designed a set of novel downstream tasks showing that both fine-tuned Nicheformer and a linear-probing model trained on the Nicheformer embedding systematically outperform existing foundation models, specifically Geneformer22, scGPT21 and UCE23 pretrained on dissociated data alone, foundation models trained in spatial data, specifically CellPLM29, and embedding models like scVI17 and principal-component analysis (PCA) for these tasks. Notes:
HV-018 FORMAL VALIDATION Cellfm-datasets: A Unified Data Infrastructure for Single-Cell and Spatial Transcriptomics Foundation Model Pretraining 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.11.731508 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Cell foundation models transfer the pretraining paradigm from language and vision to tran-scriptomic profiles. Representative systems include scBERT [8], Geneformer [9], scGPT [10], scFoundation [11], CellFM [12], and Nicheformer [13]. Notes:

HV-014 FORMAL VALIDATION Residual-stream geometry of single-cell foundation models carries incremental gene-regulatory signal across tissues 2026 BMC Bioinformatics DOI: 10.1186/s12859-026-06538-5 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We systematically investigated residual-stream geometry in scGPT and Geneformer across four tissue contexts from the Tabula Sapiens atlas, evaluating whether geometric proximity between gene vectors provides incremental predictive value for curated TRRUST transcription factor–target edges beyond expression confounds. Notes:
HV-004 FORMAL VALIDATION Discovering Candidate Anti-Aging Perturbations Using a Foundation Model for Gene Expression 2025 International Journal of Molecular Sciences DOI: 10.3390/ijms262411977 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Yes Supporting quote + locator:Our findings demonstrate that scGPT can capture aging-related transcriptional patterns and enable systematic in silico perturbation analysis at the gene level. Notes:
HV-039 FORMAL VALIDATION scGNN+: Adapting ChatGPT for Seamless Tutorial and Code Optimization 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.09.30.615735 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:In bioinformatics, these models have been applied to single-cell RNA sequencing (scRNA-seq) and high-throughput sequencing data, primarily focusing on data-centric advantages such as enhanced comprehensive analysis in accuracy and scalability2-5. Notes:

HV-034 FORMAL VALIDATION GET: a foundation model of transcription across human cell types 2023 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2023.09.24.559168 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:For example, recent developments in single-cell foundation models such as Geneformer¹⁸, scGPT¹⁹ and scFoundation²⁰ have demonstrated how diverse transcriptome profiles can be encoded in one model, enabling various contextualized downstream tasks including cell type annotation and perturbation prediction. Notes:
HV-029 FORMAL VALIDATION Elucidating the Design Space of Generative Models for Single-Cell Perturbation Prediction 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.15.732063 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:We compare against six baselines spanning continuous latent VAEs, conditional perturbation models, expression-space foundation models, and a non-parametric mean baseline: scLDM (29), scVI (18), CPA (22), scGPT (6), STATE (2), and PerturbMean. Notes:

HV-030 FORMAL VALIDATION A Foundation Model Should Not Flatten Pharmaceutical Evidence across Molecules, Sequences, Omics, Images, and Scientific Text 2024 Pharmacophore DOI: 10.51847/rbfe0o1wzp Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Protein-structure prediction depends on architectures and priors suited to sequence, evolutionary, and geometric information; transcriptomic transfer learning depends on cell-state and network contexts; and single-cell foundation modeling uses gene- and cell-specific objectives to support multi-omic, perturbational, and annotation tasks [17-19]. Notes:
HV-010 FORMAL VALIDATION evoCancerGPT: Generating Zero-Shot Single-Cell and Single-Sample Cancer Progression Through Transfer Learning 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.12.705621 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:To benchmark evoCancerGPT against alternative approaches, we compared it with a linear base-line and an scGPT embedding baseline [Cui et al., 2024] using binned MAE (bMAE) and binned Pearson correlation (bPearson), computed on the top 100 pseudotime-associated genes (see Appendix C for details). Notes:
HV-019 FORMAL VALIDATION Toward automated and explainable high-throughput perturbation analysis in single cells 2025 Patterns DOI: 10.1016/j.patter.2025.101228 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:These models can predict the effects of chemical and genetic mutations, offering computational insights into biological processes.7 Notes:

HV-047 FORMAL VALIDATION A technical review of multi-omics data integration methods: from classical statistical to deep generative approaches 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf355 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:Similarly, scGPT [180], a generative pre-trained transformer trained on over 33 million single-cell transcriptomic profiles, supports batch correction, perturbation modelling, gene network inference, and is designed to extend toward multi-omics integration. Notes:
HV-032 FORMAL VALIDATION OCOO-T : A SIMPLE AND SCALABLE VIRTUAL CELL MODEL FOR TRANSCRIPTIONAL PERTURBATION RESPONSE PREDICTION 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.08.731000 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:scGPT [1] pre-trains a generative Transformer on over 33 million human cells with a novel autoregressive gene-masking scheme, demonstrating strong transfer learning across diverse single-cell tasks. Notes:
HV-048 FORMAL VALIDATION scBaseCount: an AI agent-curated, uniformly processed, and autonomously updated single cell data repository 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.02.27.640494 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:No Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:These models are designed to capture context-dependent cellular function and behavior and predict cell state in experimentally unobserved settings such as in response to perturbation (Lopez et al., 2018; Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Rosen et al., 2023; Pearce et al., 2025; Adduri et al., 2025). Notes:

HV-036 FORMAL VALIDATION CART-GPT: A T Cell-Informed AI Linguistic Framework for Interpreting Neurotoxicity and Therapeutic Outcomes in CAR-T Therapy 2025 DOI: 10.1101/2025.08.08.669387 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Extend Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator:For model development, we applied transfer learning to adapt and fine-tune a foundation LLM model, scGPT, and further introduced a novel cell bagging strategy to identify high-impact cellular subpopulations that drive patient-level predictions. Notes:
HV-033 FORMAL VALIDATION scCCVGBen for benchmarking of single-cell representation learning anchored on a centroid-coupled variational graph attention autoencoder across scRNA-seq and scATAC-seq 2026 Frontiers in Genetics DOI: 10.3389/fgene.2026.1822168 Article page Full text 1	Review status: LOCKED Citation validity:Yes Method executed:Yes Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator:scGPT (Cui et al., 2024) and scFoundation (Hao et al., 2024) are pre-trained on tens of millions of cells and exposed via inference paths that are atlas-scoped rather than per-dataset. Rather than re-fitting them per cohort dataset (which conflicts with the pre-training contract), we evaluated their published checkpoints in the same controlled, per-dataset paired protocol on three reference datasets, including a mouse cohort whose gene symbols are upper-cased to expose orthologs to the human vocabulary scGPT checkpoint (lifting the vocabulary match from 17 to 15,935 of 31,053 genes on GSE226131). Notes: